

Stream-T1: Test-Time Scaling for Streaming Video Generation

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Research Questions

- How can test-time scaling be efficiently applied to streaming video generation to overcome the computational bottlenecks of existing diffusion-based TTS methods?
- How can temporal consistency and memory management be optimized during streaming video generation through reward-guided dynamic context routing?
- How can initial noise propagation between video chunks be leveraged to improve temporal coherence in autoregressive video generation?

Core Idea

Existing test-time scaling methods for video generation require denoising entire videos simultaneously, creating massive computational costs and preventing fine-grained temporal control. Stream-T1 shifts to streaming video generation, which naturally operates chunk-by-chunk with few denoising steps, making it ideal for TTS. The key insight is to actively optimize the generation process through three components: (1) propagating high-quality noise from previous chunks to guide current generation, (2) balancing local spatial quality with global temporal coherence through dual reward evaluation, and (3) dynamically routing memory based on semantic boundaries to preserve long-term consistency while avoiding feature corruption.

Key Formula

Stream-Scaled Noise Propagation

$$x_T^n = \beta x_T^{n-1} + \sqrt{1 - \beta^2} \epsilon, \quad \epsilon \sim N(0, I)$$

x_T^n is the initial noise for chunk n , x_T^{n-1} is the optimal noise from previous chunk, $\beta \in (-1, 1)$ controls temporal correlation strength, ϵ is fresh Gaussian noise. This maintains the marginal Gaussian distribution while establishing temporal dependency between chunks.

Dynamic Reward Fusion

$S_{\text{final}}^n = \begin{cases} \frac{n}{N} \cdot S_{\text{short}}^n + (1 - \frac{n}{N}) \cdot S_{\text{long}}^n, & \frac{n}{N} \leq \tau \\ \tau \cdot S_{\text{short}}^n + (1 - \tau) \cdot S_{\text{long}}^n, & \frac{n}{N} > \tau \end{cases}$

S_{final}^n is the combined reward for chunk n , S_{short}^n is spatial quality score, S_{long}^n is temporal coherence score, n/N is chunk position ratio, τ is threshold constraint. This balances local aesthetics and global coherence with increasing weight on spatial quality over time, capped by threshold τ .

Key Figure

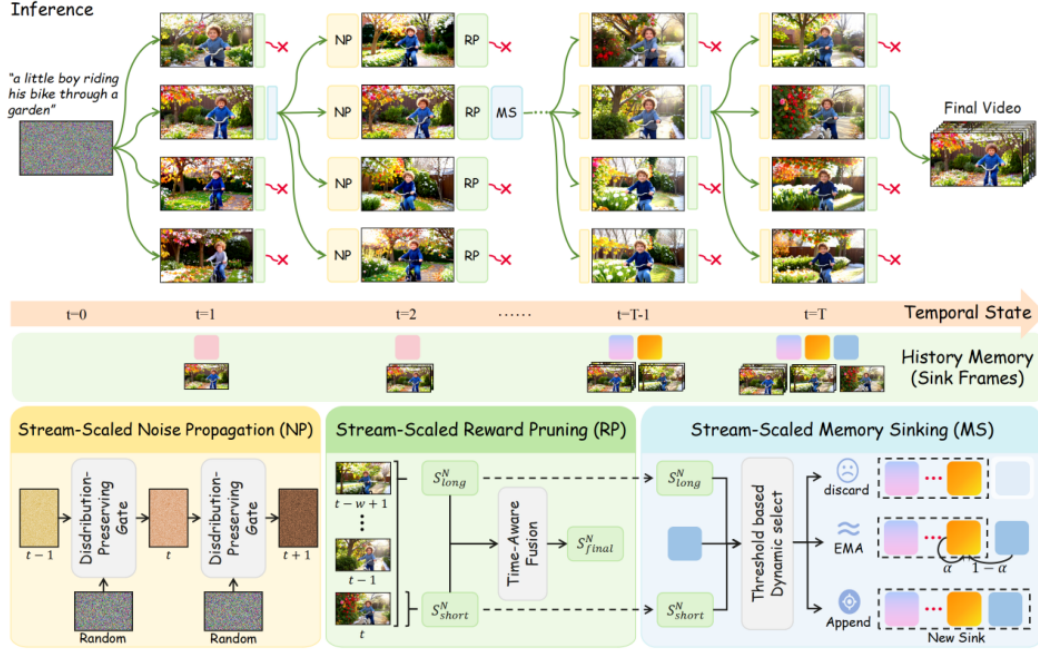


Figure 1 Overview of the Stream-T1 chunk-level scaling pipeline. For each chunk, our test-time scaling mechanism operates in three sequential stages: (1)Stream-Scaled Noise Propagation, where the initial noise latents for the current chunk are actively sampled, conditioned on the optimal noise trajectories retained from the previous chunk. (2)Stream-Scaled Reward Pruning, where fully generated chunk candidates are holistically evaluated via a Long-Short Combined Reward to balance local spatial aesthetics with global temporal coherence. Based on this feedback, suboptimal branches are pruned to propagate only the optimal generation trajectories.(3)Stream-Scaled Memory Sinking, where the memory sink is adaptively updated based on reward scores to preserve long-term semantics and guide the subsequent video generation. We present the full search trajectory of the illustrated case in the Figure 2.

Figure 1: Shows the complete Stream-T1 pipeline with all three components operating sequentially on chunk generation, illustrating the noise propagation, reward pruning, and memory sinking mechanisms.

Limitations

- Requires careful tuning of multiple hyperparameters (β for noise propagation, τ for reward threshold, sliding window sizes)
- Semantic boundary detection may fail during gradual scene transitions, leading to suboptimal memory routing decisions
- Performance depends heavily on quality of reward models used for short-term and long-term evaluation
- Computational overhead still increases with beam search width and candidate generation, though more efficient than full-video TTS methods